

AI Study Assistant - Backend Folder Guide

Purpose of Each Folder/File

routes

Defines API endpoints (URLs). Example: POST /api/auth/register, POST /api/auth/login. Think of routes as the front door to the backend.

controllers

Contains the business logic for each route, e.g., registerUser(), loginUser(). Keeps code out of index.js.

middleware

Runs before/after requests. Used for authentication, JWT verification, logging, error handling.

models

Defines database structure (User, PDF, etc.). Used with MongoDB/Mongoose.

config

Stores configuration code such as MongoDB connection and environment setup.

index.js

Application entry point. Creates the Express app, registers middleware and routes, and starts the server.

package.json

Project configuration: dependencies, scripts, metadata.

.env

Stores secrets and configuration like PORT, MONGO_URI, JWT_SECRET, GEMINI_API_KEY. Never commit this file.

Typical Request Flow

React Frontend → Route → Controller → Model (MongoDB) → Controller → JSON Response → React